

ETHCF Swapboard Audit Summary

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Version 2.0

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March 4, 2026

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for ETHCF on Swapboard. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated March 4, 2026, was delivered to ETHCF, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 1 days, the Cyfrin team conducted an audit on the Swapboard code provided by ETHCF. The findings consist of 2 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Swapboard
Repository	swapboard
Commit	0c0a7eea363f...
Fix Commit	82f9b4902ae5...
Audit Timeline	Feb 25th - Feb 25th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	2	1	1
Informational	3	2	1
Gas Optimizations	6	6	0
Total	11	9	2

6 Audit Scope

The audit scope was limited to:

```
contracts/src/Swapboard.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

`Swapboard` is a trustless OTC bulletin board for ERC20 token swaps on Ethereum. It implements a simple on-chain orderbook where makers deposit tokens into escrow and takers atomically fill orders by providing the requested counterpart tokens.

Key design properties:

- **Atomic all-or-nothing fills:** orders cannot be partially filled; each fill transfers the full amounts between maker and taker in a single transaction
- **Maker-escrowed `tokenA`:** when creating an order, the maker's selling token (`tokenA`) is transferred into the contract and held in escrow until the order is filled or cancelled
- **Direct taker-to-maker `tokenB`:** when filling an order, the buying token (`tokenB`) is transferred directly from the taker to the maker without passing through the contract
- **Native ETH convenience wrappers:** the contract provides `createOrderWithEth`, `fillOrderWithEth`, `cancelOrderUnwrap`, and `fillOrderUnwrap` functions that automatically wrap/unwrap ETH via the canonical `WETH` contract, allowing users to trade native ETH without manually interacting with `WETH`
- **Order lifecycle:** each order transitions from active to inactive exactly once, either through filling or cancellation

9 Attestation

Cyfrin attests that it performed a security review of the Swapboard code provided by ETHCF.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-03-04-cyfrin-ethcf-swapboard-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to ETHCF has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=447c8be7318971a369a9ca1a757aee171fd28b4fef811c2e98b158b3caea0629
F=2026-03-04-cyfrin-ethcf-swapboard-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.